Unit 12, Lesson 3: Solving Problems With Secant and Tangent Segments

Benchmark

MA.912.GR.6.1 Solve mathematical and real-world problems involving the length of a secant, tangent, segment, or chord in a given circle.

Learning Target

- ☐ I can solve mathematical and real-world problems that involve lengths formed by the intersection of a secant and a tangent segment.

12.3.1 Warm-Up: Notice and Wonder

1. The Seminole Native American tribe is known for their novel and colorful dress consisting of patchwork. The decorative technique of patchwork was developed by Seminole women before 1920. Patchwork was rapidly adopted to further embellish the already colorful clothing. As time went on, the designs became more intricate as the seamstresses became more skilled.

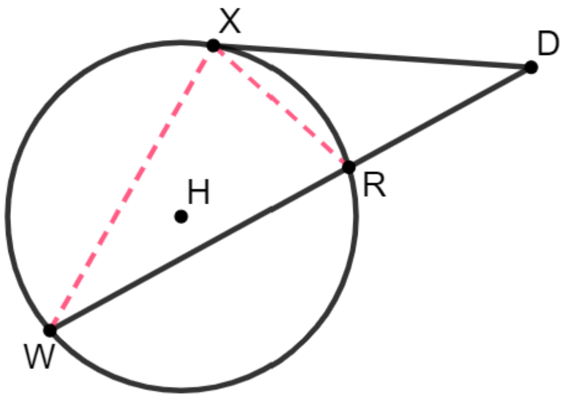
   a. What do you notice about the patchwork?

   b. What do you wonder?

12.3.2 Exploration: Lengths of Intersecting Secant and Tangent Segments

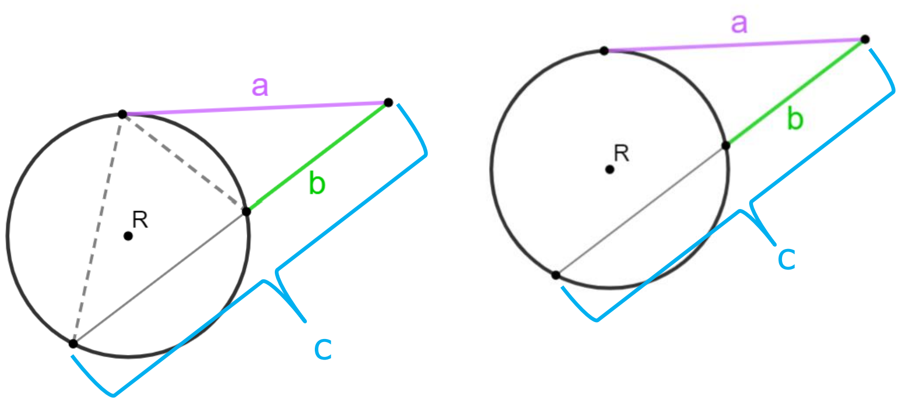
Work with your partner to complete the following.

1. Circle H has points X, R, and W on the circle. Secant $\overline{WD}$ and tangent $\overline{XD}$ intersect at point D outside the circle. Given that $\Delta DWX \sim \Delta DXR$, $XD = 15$, and $RD = 9$.



   a. Use the similiar triangles to write a proportional relationship that can be used to determine the length of $\overline{RW}$.

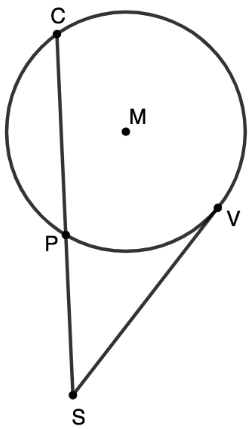
   b. Use the diagrams of circle R to write a formula that can be used to find the segment lengths of the intersecting secant and tangent lines.



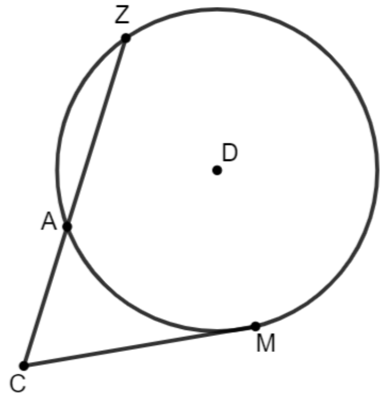
If a secant and tangent segment are drawn to a circle from the same external point, then the square of the measure of the tangent segment is equal to the product of the measures of the secant segment and its external secant segment.

12.3.3 Guided Practice: Solving Problems Involving Intersecting Secant and Tangent Segments

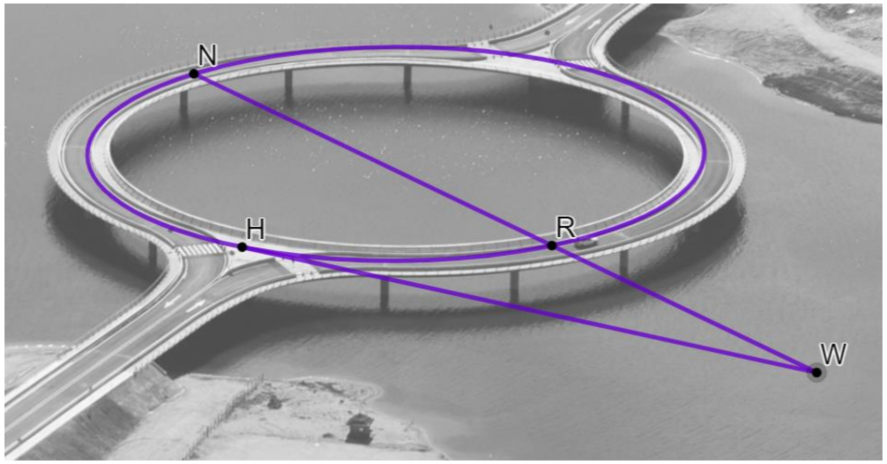
1. Circle M has points C, V, and P on the circle. Secant $\overline{CS}$ and tangent $\overline{VS}$ intersect at point S outside the circle. $CP = 4$ and $SP = 3$. What is the length of $\overline{SV}$? Round to the nearest thousandth.



2. Circle D is shown with points Z, A, and M on the circle. Secant $\overline{CZ}$ and tangent $\overline{CM}$ intersect at point C outside the circle. $AZ = 16$, $AC = (x - 3)$, and $MC = (x + 3)$. What is the length of $\overline{CM}$?

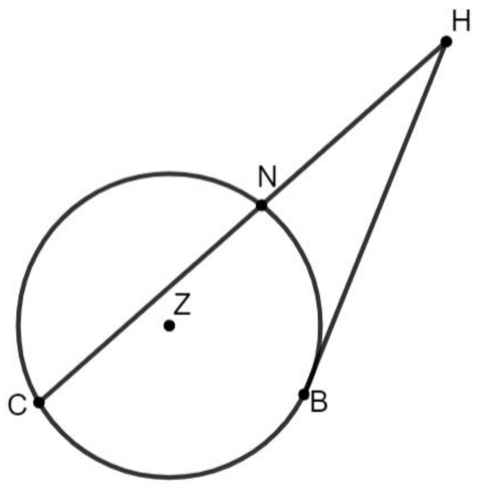


3. A drone flew over the Laguna Garzon Bridge in Uruguay and took pictures at three points on the bridge: N, R, and H. Tangent $\overline{HW}$ and secant $\overline{NW}$ intersect at point W off the bridge. If $HW = 280$ feet (ft) and $RW = 160$ ft, what is the length of $\overline{NW}$?

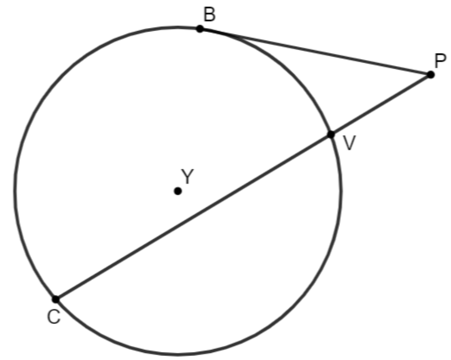


12.3.4 Your Turn!

1. Circle Z is shown with points C, N, and B on the circle. Secant $\overline{CH}$ and tangent $\overline{BH}$ intersect at point H outside the circle. $CN = (7x - 6)$, $NH = 12$, and $HB = 18$. Find the value of x.



2. Isabella and Gregory were working on finding the length of $\overline{BP}$. Circle Y is given, where B, V, and C are points on the circle. Secant $\overline{CP}$ and tangent $\overline{BP}$ intersect at point P outside the circle. $CV = 12$ and $VP = 4$.



Isabella's and Gregory's work to find the length of $\overline{BP}$ is shown.

Isabella's Work	Gregory's Work
$x^2 = 4(4+12)$ $x^2 = 4(16)$ $x^2 = 64$ $x = 8$	$x^2 = 12(4+12)$ $x^2 = 12(16)$ $x^2 = 192$ $x = 13.9$

a. Whose work do you agree with?

b. Write a note to the person whose work is incorrect explaining their mistake.

12.3.5 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.

Unit 12, Lesson 4: Solving Problems With Lengths of Two Tangents

Benchmark

MA.912.GR.6.1 Solve mathematical and real-world problems involving the length of a secant, tangent, segment, or chord in a given circle.

Learning Target

- ☐ I can solve mathematical and real-world problems that involve lengths formed by the intersection of two tangents.